Wenbo Zeng

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EDUCATION

B.E. in Zhejiang University Sept. 2023 - Present
Edison Honors Class, College of Electrical Engineering, Zhejiang University, Hangzhou, China
Major: Electrical Engineering and its Automation
GPA: 4.19/4.3 & 92.79/100

Core Module
Power System Optimization (100), Intelligent Control and Applications (98), Numerical Analysis (98), Electrical Network Analysis (96), Control Theory (B) (95), High Voltage Engineering (95), Power System Transient Analysis (95), Protective Relaying and Automatic Devices (95), Power Electronics (93), Electric Circuits and Electronics II (93), Power Information Technology (93), Complex Variable Functions and Integral Transforms (93), Operations Research (92)

RESEARCH INTERESTS

Edge Intelligence in Smart Grids, Data Analytics for Power Systems

RESEARCH FUNDING

F1 Future Academic Star for Undergraduates. 2025
RMB 20,000, Zhejiang University (**30** out of all undergraduates university-wide), Principal Student Investigator.
Project: Sensorless Control and Fault Ride-Through of Doubly-Fed Induction Generators.

F2 National Innovation Key Project for Undergraduates. 2025
RMB 24,000, Zhejiang University (**5** out of all undergraduates university-wide), Principal Student Investigator.
Project: Sensorless Control and Fault Ride-Through of Doubly-Fed Induction Generators.

SCHOLARSHIPS & AWARDS

Scholarships

- National Scholarship of China (top 3.5%) (2026)
- National Scholarship of China (top 3.5%) (2025)
- National Scholarship of China (top 3.5%) (2024)
- First Prize Scholarship of Zhejiang University (top 7%) (2026, 2025, 2024)
- Panasonic ”Ikuei” Excellence Scholarship (First Class) (4 out of 200+ students) (2025)

Honors

- Top 10 Outstanding Students of Lantian College (10 out of 1700+ students) (2024)
- Excellent Student of Zhejiang University (2026, 2025, 2024)
- Distinguished Academic Award of Zhejiang University (2026, 2025, 2024)
- Distinguished Innovation and Entrepreneurship Award of Zhejiang University (2026, 2025, 2024)
- Distinguished Social Work Award of Zhejiang University (2025)

Contest Awards

- Zhejiang Provincial Mathematics Competition for Undergraduate Students — First Prize (2024)
- Zhejiang Provincial Physics Competition for Undergraduate Students — Second Prize (2024)
- National English Competition for College Students — Third Prize (2024, 2025)

RESEARCH PUBLICATIONS

Journal Articles

JCR quartiles: Engineering, Electrical & Electronic; impact factors: 2025 JCR.

J1 **Wenbo Zeng**, Tao Wang, Yuhao Wang, Han Wu, Xiang Meng, Chen Zhao, Zhenxiao Zhang and Lijian Wu. Sensorless Control of Doubly-Fed Induction Generators with Grid Fault Ride-Through Capability. *IEEE Transactions on Energy Conversion* (JCR Q1, IF=6.1), **Accepted**.

J2 Qi Li, **Wenbo Zeng**, Xueyuan Cui and Yi Wang. Uncertainty Allocation-based Tube Model Predictive Control for Building Energy Management. *IEEE Transactions on Smart Grid* (JCR Q1, IF=10.1), **Accepted**.

J3 Haoyuan Deng, Lingyu Chen, **Wenbo Zeng**, Xueyuan Cui and Yi Wang. Federated Heterogeneous-agent Reinforcement Learning for the Dispatch of Energy Hubs. *IEEE Transactions on Smart Grid* (JCR Q1, IF=10.1), **Accepted**.

Patent

P1 Tao Wang, **Wenbo Zeng** and Lijian Wu. Sensorless Control Method and System for Doubly-Fed Induction Wind Generator. ZL 202510535050.8, **Granted**; first student inventor.

P2 Tao Wang, **Wenbo Zeng**, Lijian Wu, Chen Zhao and Zhenxiao Zhang. Rotor Position Estimation Method for Doubly Fed Induction Generators with Large-Signal Stability. Application No. 202610808675.1, **Under Substantive Examination**; first student inventor.

RESEARCH EXPERIENCE

Zhejiang University-Shanghai Electric Wind Power R&D Center Oct. 2024 – Present

Undergraduate Research Assistant, supervised by Prof. Tao Wang

Topic 1: Sensorless Control and Fault Ride-Through of Doubly-Fed Induction Generators

- Developed sensorless DFIG control with fault ride-through under up to 80% voltage dips; first-authored a paper accepted by *IEEE Transactions on Energy Conversion* (J1)
- Secured research funding of RMB 44,000 (F1, F2)
- Filed two invention patents as the first student inventor (P1–P2)

The University of Hong Kong Nov. 2025 - Present

Research Collaboration, supervised by Prof. Yi Wang

Topic 1: Uncertainty Allocation-based Tube MPC for Building Energy Management

- Implemented deterministic, robust, and stochastic MPC baselines; processed experimental data and analyzed comparative results
- Co-authored a paper accepted by *IEEE Transactions on Smart Grid* as second author (J2)

Topic 2: Federated Heterogeneous-agent Reinforcement Learning for Energy Hubs

- Implemented reinforcement learning and optimization baselines for energy hub dispatch; processed experimental data and analyzed comparative results
- Co-authored a paper accepted by *IEEE Transactions on Smart Grid* as third author (J3)

English Proficiency

TOEFL: 112/120 (R:30, L:30, W:27, S:25)
GRE: 328/340 + 3.5/6.0 (V:160, Q:168, AW:3.5)

SKILLS

Proficient: Matlab/Simulink, C, Python